

In the grading of the problems special attention will be paid to check that answers are substantiated and that the procedure of calculations is well documented. Your solution to the problem set can be written in Danish or English as you prefer.

Problem 1

Consider a causal system given by the following system function

$$H(z) = 1 + 3.5z^{-1} + 1.5z^{-2}$$

and assume that the causal system is used as a model for a communication channel that will distort signals passing through it.

1. Design an equalizer that can mitigate the distortion.
2. Make a time-domain simulation of the system and the equalizer using a test signal of your choice. Explain the choice of test signal and procedures, present plots, and analyze the performance of the equalizer.

Next, assume that after the signal has passed through the communication channel is it further distorted by additive white Gaussian noise before it enters the equalizer designed above.

3. How are the results affected by the additive noise?

Problem 2

Consider the random process $x[n]$ created as $x[n] = 4w[n] - 2w[n - 3]$ where $w[n]$ is unit variance white Gaussian noise.

1. Compute an analytic expression for the autocorrelation $r_{xx}[l]$.
2. Plot the power spectral density of the random process. What can be learned from the plot?

Assume $x[n]$ is filtered through a FIR filter with $H(z) = 5 + z^{-1}$.

3. Compute the analytic expressions for $r_{yx}[l]$ and $r_{yy}[l]$.

Problem 3

Let a signal and interpolation kernel be given by

$$x[n] = \begin{cases} 2 & n = 0 \\ 1 & n = 1 \\ 0 & \text{elsewhere} \end{cases} \quad \text{and} \quad g[n] = \begin{cases} -1 & n = -1 \\ 3 & n = 0 \\ -1 & n = 1 \\ 0 & \text{elsewhere} \end{cases}$$

1. Upsample $x[n]$ by a factor of 3 and sketch the upsampled signal.
2. Compute the interpolated values $x_I[1]$ and $x_I[2]$ using first linear interpolation and second the proposed $g[n]$ kernel.
3. Discuss the performance of the $g[n]$ interpolation kernel relative to ideal and linear interpolation.

Problem 4

The file `SampleRateData.dat` contains a short segment of a signal that is sampled at 1 kHz. The signal, denoted by $x[n]$, can be loaded into MATLAB with `x=load('SampleRateData.dat')`. In this problem we will first form the signal $y[n]$ by decimating $x[n]$ by two. Afterwards, we create the signal $z[n]$ by 'undoing' the decimation, i.e., $z[n]$ is found by interpolating $y[n]$ by two.

1. Perform the decimation and interpolation. Document the procedure, plot the signals at each stage, and explain the shape of the signals after decimation and after interpolation.

Problem 5

Consider a two-channel filter bank where it is intended to operate it as a quadrature mirror filter with $H(z) = 1 + \frac{1}{2}z^{-1}$.

1. Are $H_0(z)$ and $H_1(z)$ power complementary?
2. Will the choice of $H(z)$ lead to an aliasing-free filter bank?
3. Discuss the choice of $H(z)$ for a quadrature mirror filter bank.

Problem 6

The file `dataset.dat` contains 2048 samples of an unknown signal. The signal can be loaded into MATLAB with `x=load('dataset.dat')`.

1. Compute and plot the autocorrelation for an appropriate range of lags. Compare it with the autocorrelation for an equal length realization of white Gaussian noise with the same power. What can be derived about the signal in the file from the autocorrelation?
2. Estimate and plot the power spectral density of the signal using first a periodogram and second Welch's method. Discuss the results and explain the pros and cons of the two methods.